LAB 05

1. Compress a file by gzip and bzip2 commands and decompress it again.
State the differences between two commands.

```
[mina_shafik@localhost ~]$ ls
list_dir  modify  mydir  mydtxt.txt  myfile  myteam  oldpasswd  snake  source_dir  source_file  target_dir  testdir  testfile
[mina_shafik@localhost ~]$ sudo gzip snake
[mina_shafik@localhost ~]$ ls
list_dir  modify  mydir  mydtxt.txt  myfile  myteam  oldpasswd  snake.gz  source_dir  source_file  target_dir  testdir  testfile
[mina_shafik@localhost ~]$ sudo gunzip snake
[mina_shafik@localhost ~]$ ls
list_dir  modify  mydir  mydtxt.txt  myfile  myteam  oldpasswd  snake  source_dir  source_file  target_dir  testdir  testfile
[mina_shafik@localhost ~]$
```

2. What is the command used to view the content of a compressed file.

sudo zcat filename.gz

3. Backup /etc directory using tar utility.

```
[mina_shafik@localhost etc]$ sudo tar -cf etc-backup.tar /etc
tar: Removing leading `/' from member names
tar: /etc/etc-backup.tar: file is the archive; not dumped
[mina_shafik@localhost etc]$ ls  /etc/
adjtime             csh.cshrc                GREP_COLORS  kernel          makedumpfile.conf.sample  pkcs11                  selinux         sysctl.d
aliases             csh.login                groff        krb5.conf       man_db.conf               pki                     services        systemd
alternatives        dbus-1                   group        krb5.conf.d     microcode_ctl             pm                      sestatus.conf   system-release
anacrontab          default                  group-       ld.so.cache     mime.types                popt.d                  shadow          system-release-cpe
audit               depmod.d                 grub2.cfg    ld.so.conf      mke2fs.conf               printcap                shadow-         terminfo
authselect          dhcp                     grub.d       ld.so.conf.d    modprobe.d                profile                 shells          tmpfiles.d
bash_completion.d   DIR_COLORS               gshadow      libaudit.conf   modules-load.d            profile.d               skel            tpm2-tss
bashrc              DIR_COLORS.lightbgcolor  gshadow-     libibverbs.d    motd                      protocols               ssh             udev
binfmt.d            dnf                      gss          libnl           motd.d                    rc.d                    ssl             vconsole.conf
chrony.conf         dracut.conf              host.conf    libreport       mtab                      rc.local                sssd            vimrc
chrony.keys         dracut.conf.d            hostname     libssh          NetworkManager            redhat-release          statetab.d      virc
cifs-utils          environment              hosts        libuser.conf    networks                  resolv.conf             subgid          wgetrc
cron.d              etc-backup.tar           httpd        locale.conf     nftables                  rocky-release           subgid-         X11
cron.daily          ethertypes               inittab      localtime       nsswitch.conf             rocky-release-upstream  subuid          xattr.conf
cron.deny           exports                  inputrc      login.defs      nsswitch.conf.bak         rpc                     subuid-         xdg
cron.hourly         filesystems              iproute2     logrotate.conf  openldap                  rpm                     sudo.conf       yum
cron.monthly        firewalld                issue        logrotate.d     opt                       rsyslog.conf            sudoers         yum.conf
crontab             fonts                    issue.d      lvm             os-release                rsyslog.d               sudoers.d       yum.repos.d
cron.weekly         fstab                    issue.net    machine-id      pam.d                     rwtab.d                 sudo-ldap.conf
crypto-policies     gcrypt                   kdump        magic           passwd                    sasl2                   sysconfig
crypttab            gnupg                    kdump.conf   mailcap         passwd-                   security                sysctl.conf
[mina_shafik@localhost etc]$
```

4. Starting from your home directory,
find all files that were modified in the last two day.

```
[mina_shafik@localhost etc]$ sudo find ~ -type f -mtime -2
/home/mina_shafik/.bash_history
/home/mina_shafik/etc-backup.tar
[mina_shafik@localhost etc]$
```

5. Starting from /etc, find files owned by root user.

```
[mina_shafik@localhost etc]$ sudo find /etc/ -user root
/etc/
/etc/mtab
/etc/fstab
/etc/crypttab
/etc/lvm
/etc/lvm/devices
/etc/lvm/devices/system.devices
/etc/lvm/archive
/etc/lvm/archive/rl_00000-899613452.vg
/etc/lvm/backup
/etc/lvm/backup/rl
/etc/lvm/cache
/etc/lvm/lvm.conf
```

6. Find all directories in your home directory.

```
[mina_shafik@localhost etc]$ sudo find ~ -type d
/home/mina_shafik
/home/mina_shafik/myteam
/home/mina_shafik/testdir
/home/mina_shafik/mydir
/home/mina_shafik/source_dir
/home/mina_shafik/target_dir
/home/mina_shafik/target_dir/source_dir
/home/mina_shafik/list_dir
[mina_shafik@localhost etc]$
```

7. Write a command to search for all files on the system that, its name is ".profile".

```
Last login: Sun Mar 15 17:36:59 2026
[root@localhost ~]# sudo find / -type f -name ".profile"
find: '/proc/62257': No such file or directory
[root@localhost ~]#
```

8. Identify the file types of the following:
/etc/passwd, /dev/pts/0, /etc, /dev/sda

```
[root@localhost ~]# file /etc/passwd /dev/pts/0 /etc /dev/sda
/etc/passwd: ASCII text
/dev/pts/0:  character special (136/0)
/etc:        directory
/dev/sda:    block special (8/0)
[root@localhost ~]#
```

9. List the inode numbers of /, /etc, /etc/hosts

```
[root@localhost ~]# ls -id / /etc /etc/hosts
33575041 drwxr-xr-x. 78 root root 8192 Mar 15 17:34 /etc
     128 dr-xr-xr-x. 25 root root 4096 Mar  7 01:09 /
33794713 -rw-r--r--.  1 root root  158 Jun 23  2020 /etc/hosts
[root@localhost ~]#
```